

2018-Ph-H1-Q22

Section: Electricity

Topic: Current, PD, Power, Resistance

Two resistors of 6Ω and 3Ω are connected in parallel.

What is their combined resistance?

Solution:

Use:

$$\frac{1}{R} = \frac{1}{6} + \frac{1}{3} = \frac{1+2}{6} = \frac{3}{6} \Rightarrow R = 2\Omega$$

Answer: 2Ω

Revision Tip:

Parallel resistors: always use reciprocal formula. Total resistance in parallel is **less than the smallest resistor**.